



POLITÉCNICA



Large-Scale Smart Grids as System of Systems

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Introduction

▶ A System of Systems

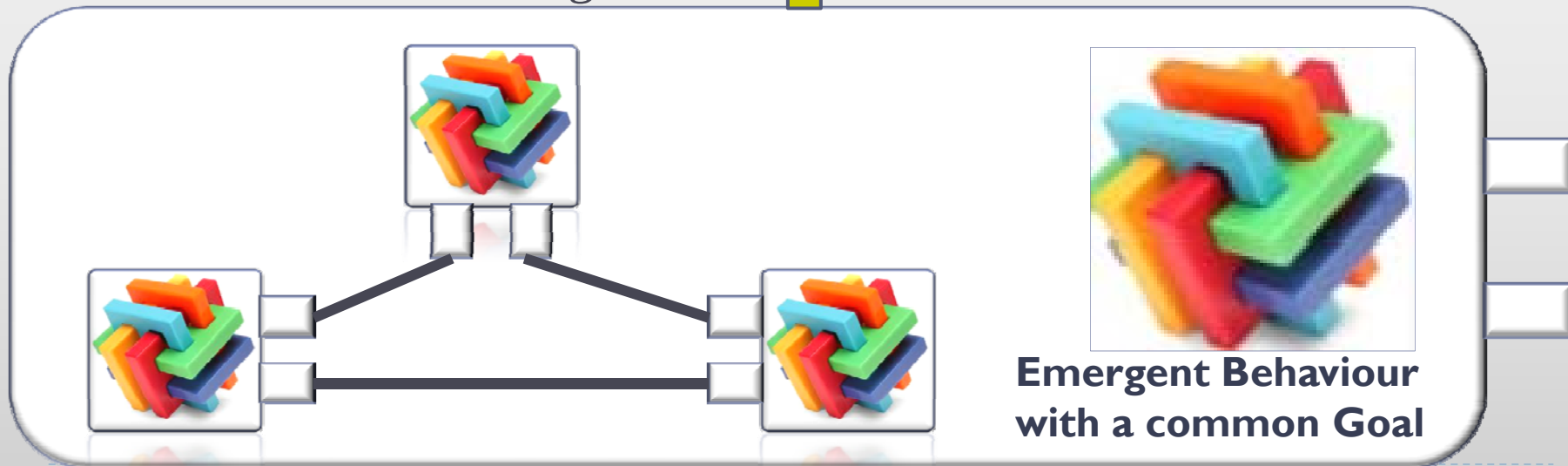
- ▶ Conceptualized = a kind of systems which are built from components



- ▶ Large scale systems in their own right
- ▶ Independent
- ▶ Complex
- ▶ Heterogeneous



An Example: Smart Grids





Introduction

- ▶ Smart Grids as a SoS → From Smart Grids To Large-Scale Smart-Grids

- ▶ Challenges:

- ▶ Conceptualize Large-Scale Smart-Grids as a SoS
- ▶ Architectural Design
- ▶ Guarantee Interoperability



An Architectural Framework for Large-Scale Smart Grids



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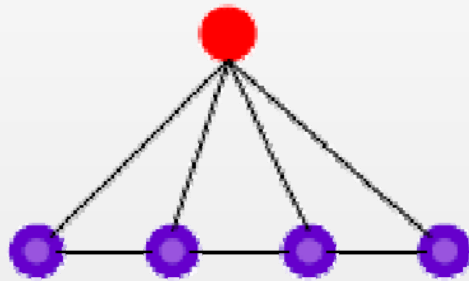


SoS

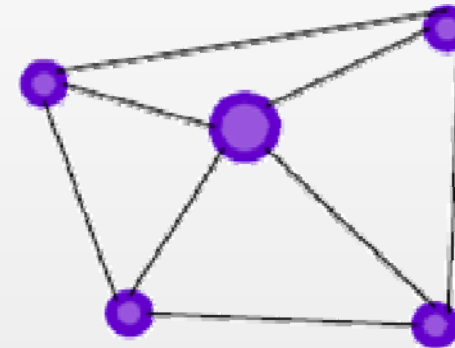
- ▶ Characterized by 5+3 criteria
 - ▶ Maier's 5 criteria
 - ▶ Operational independence of elements
 - ▶ Managerial independence of elements
 - ▶ Evolutionary development
 - ▶ Emergent behavior
 - ▶ Geographic distribution
 - ▶ DeLaurentis' 3 criteria
 - ▶ Inter-disciplinary
 - ▶ Heterogeneous
 - ▶ Network of systems.

SoS

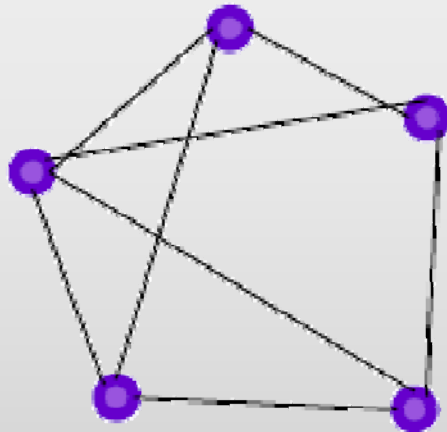
► Classification *[Maier] [Dahmann et al]*



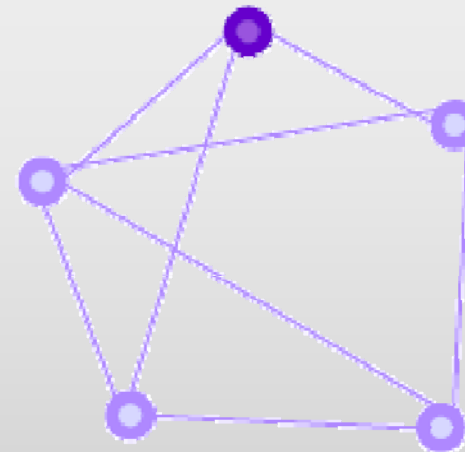
Directed



Acknowledged



Collaborative



Virtual



SoS

- ▶ Software Architecture
 - ▶ Should be:
 - ▶ Open
 - ▶ Loosely coupled
 - ▶ Welcome to changes
 - ▶ Guarantee the interoperability
 - ▶ Should define stable intermediate forms
 - ▶ Should use Standards



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Smart Grids

- ▶ Smart Grids are composed of a broad range of energy resources, from large generating systems to smaller generating systems, all of them operating as a single system providing both power and heat to their consumers





Smart Grids

- ▶ Definitions:
 - ▶ **Smart Grid:**
 - ▶ A set of microgrids, energy resources and systems, and consumers that are connected to the main grid
 - ▶ **Microgrid:**
 - ▶ Small-scale grid that is designed to provide electrical and/or thermal energy for local loads and communities
 - ▶ **Large-Scale Smart Grid:**
 - ▶ Smart Grid that scales up the electricity service to millions of customers
- ▶ Promote the **integration** of traditional and renewable energy resources in **distributed, open, and self-managed** way.
- ▶ Mean a shift from current centralized **energy infrastructures** towards more **distributed** ones

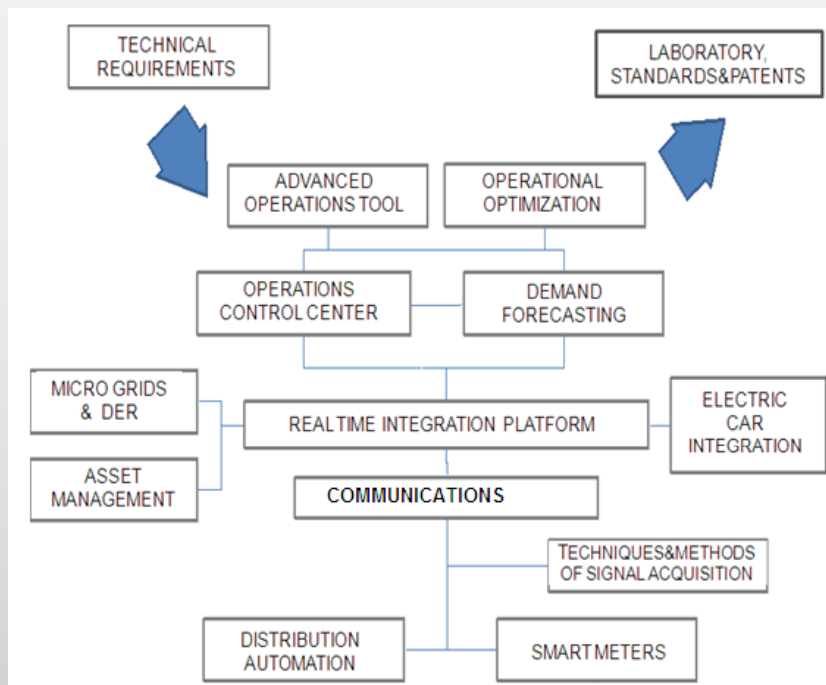


Smart Grids

- ▶ Require innovative models and software architectures for:
 - ▶ Integrating the **distributed renewal** energy resources
 - ▶ Providing energy infrastructure with **intelligent** communication, monitoring, control, and management
 - ▶ Enabling **two-way exchange** of power and information between suppliers and consumers (or prosumers)
 - ▶ Managing the growing energy **demand**

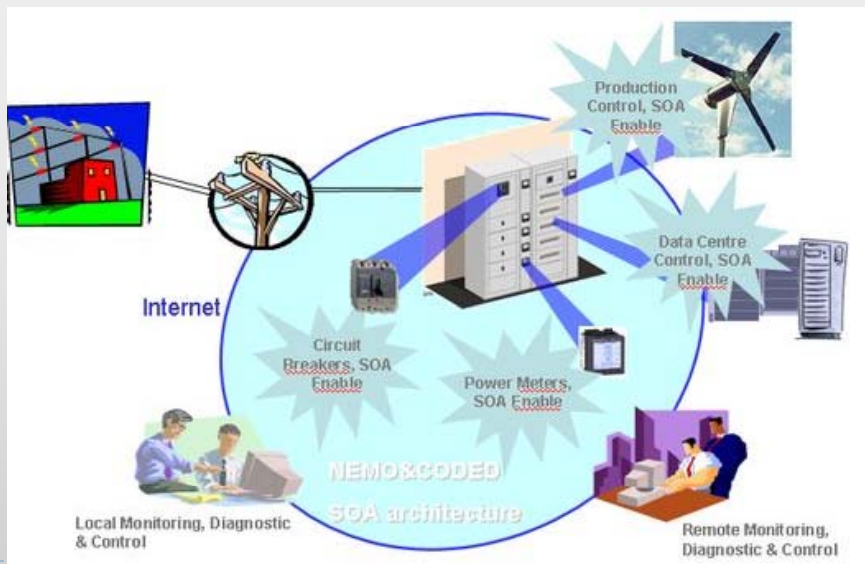
Smart Grids

- ▶ **ENERGOS:** Technologies for automated and intelligent management of the future power distribution networks
 - ▶ **Goal:** to acquire technical capabilities on basic technologies for Smart Grids
 - ▶ **Outcomes:** Middleware technologies and uses case from the Energy Domain



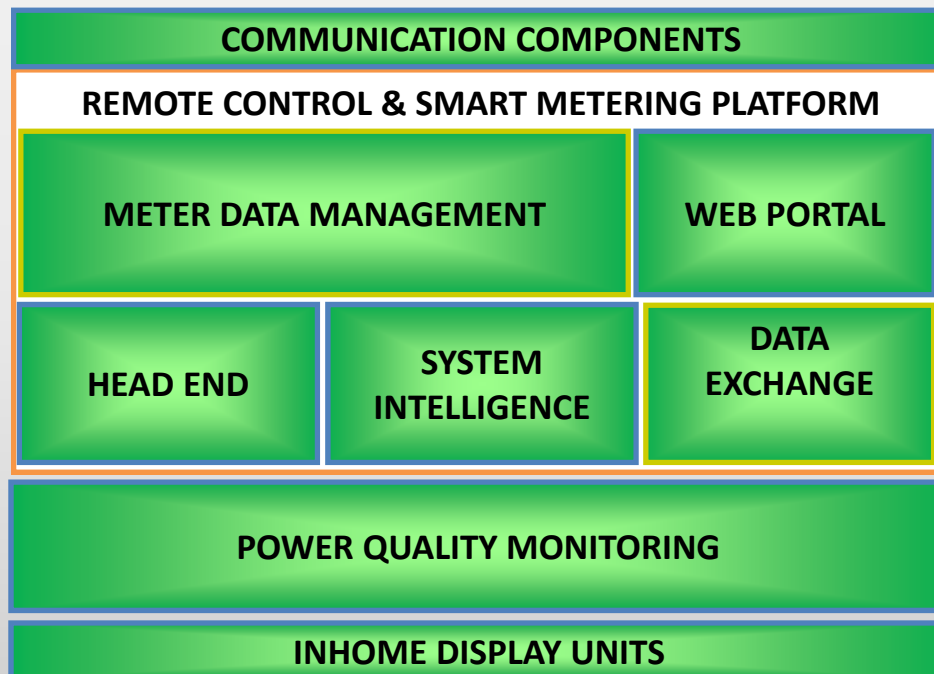
Smart Grids

- ▶ **NEMO&CODED: NEtwork MOnitoring & COntrol, Diagnostic for Electrical Distribution**
 - ▶ **Goal:** Monitoring and control technologies for electricity networks, focusing on the application of current Web Services technologies into the energy domain
 - ▶ **Outcomes:** Head-End Integration, Use of DPWS Standards, Acquisition Metering Platform



Smart Grids

- ▶ **IMPONET: Intelligent Monitoring of Power Networks**
 - ▶ **Goal:** To develop an advanced platform for future Smart Metering
 - ▶ **Outcomes:** Modules of the Advanced Metering Platform, Advanced Processing and Storing Infrastructure, Integration of MDM with Nemo Connector via Web Services and with Real-Time Nodes via XTPP





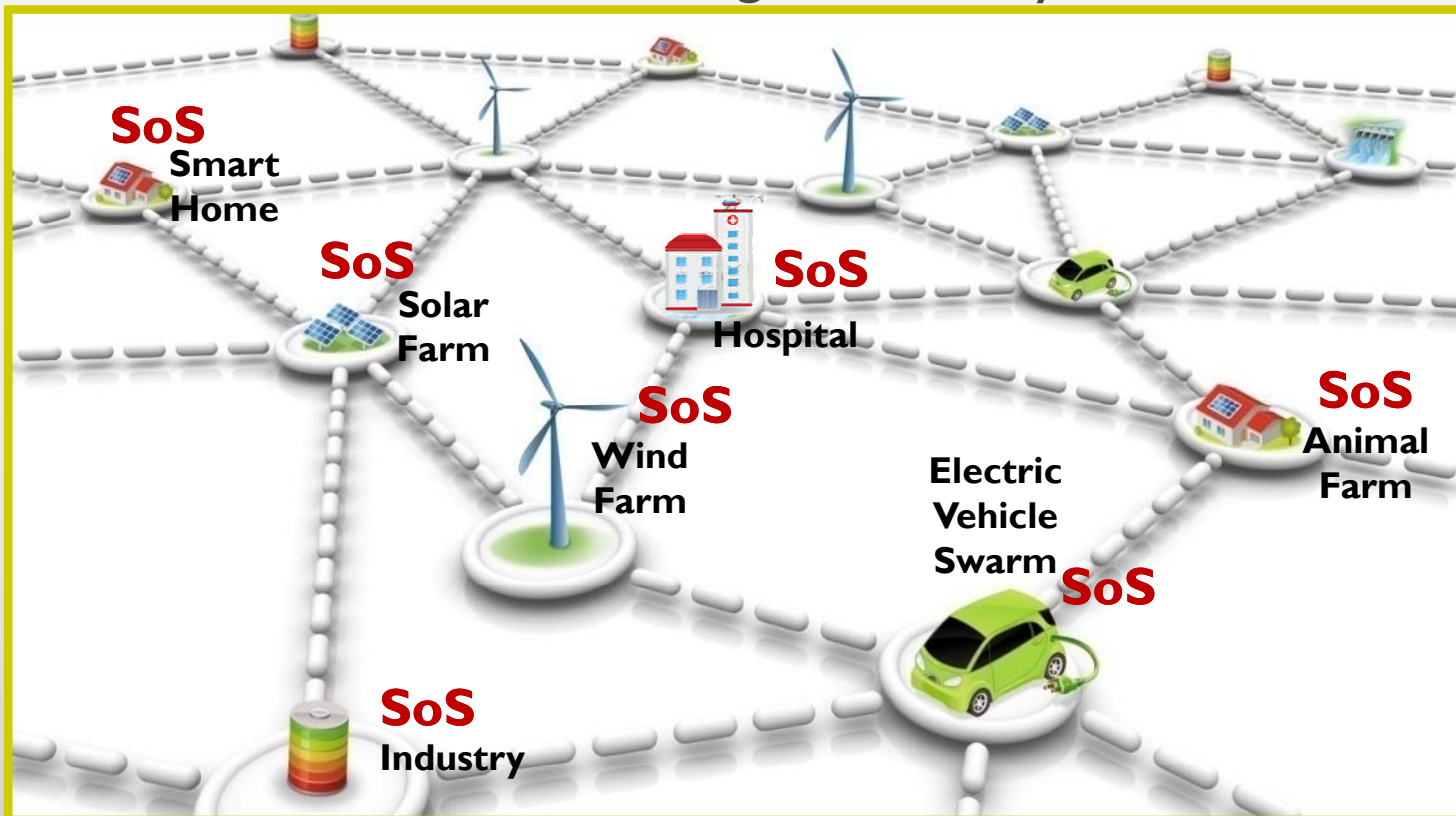
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Large-Scale Smart Grids as SoS

► Conceptualization

- The **Large-Scale Smart Grids** are constituted by resources interconnected among them. Many of these resources are systems



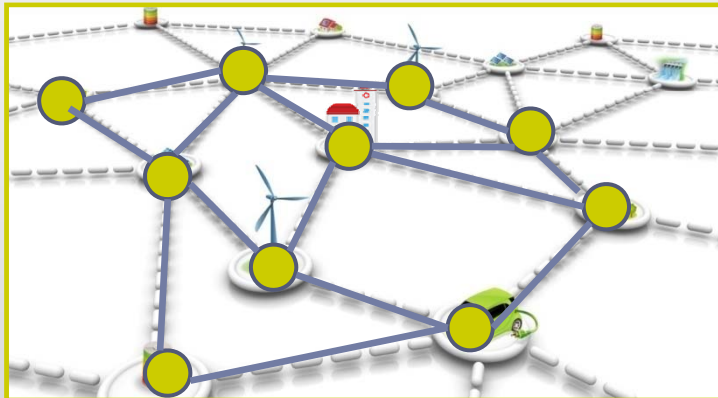
A SoS of SoS

Large-Scale Smart Grids as SoS

► Conceptualization

► The SoS that compose the Large-Scale Smart Grid have:

- Their own management goals and business purposes and work on their own and *collaboratively* with the rest of SoS
- An agreement, usually with the utility, in terms of power decisions, i.e. power supply, billing, power sale, power storage, etc



All of them have a
agreement with
the Utility



Collaborative / Acknowledge

Large-Scale Smart Grids as SoS

► Conceptualization

► Characterization criteria:

- Operational independence of elements
- Managerial independence of elements
- Evolutionary development
- Emergent behavior
- Geographic distribution
- Network of systems
- Inter-disciplinary
- Heterogeneous

Smart Grid's SoS work on their own

Smart Grid's SoS have their own management goals and business purposes

Smart Grids have an open nature, and a flexible and scalable infrastructure

Smart Grids leverage emergent behavior and requires run-time capabilities

Smart Grid services are decentralized and its infrastructure is distributed

Smart Grid are composed of systems such as solar farms, power plants, hospitals, wind farms, animal farms, industries, the swarm of electric vehicles, smart homes, etc.





Large-Scale Smart Grids as SoS

► Conceptualization

► Software Architecture

- ☐ Open, Loosely coupled, Welcome to changes, Guarantee the interoperability
- Should define stable intermediate forms
- Should use Standards



► Real time capabilities

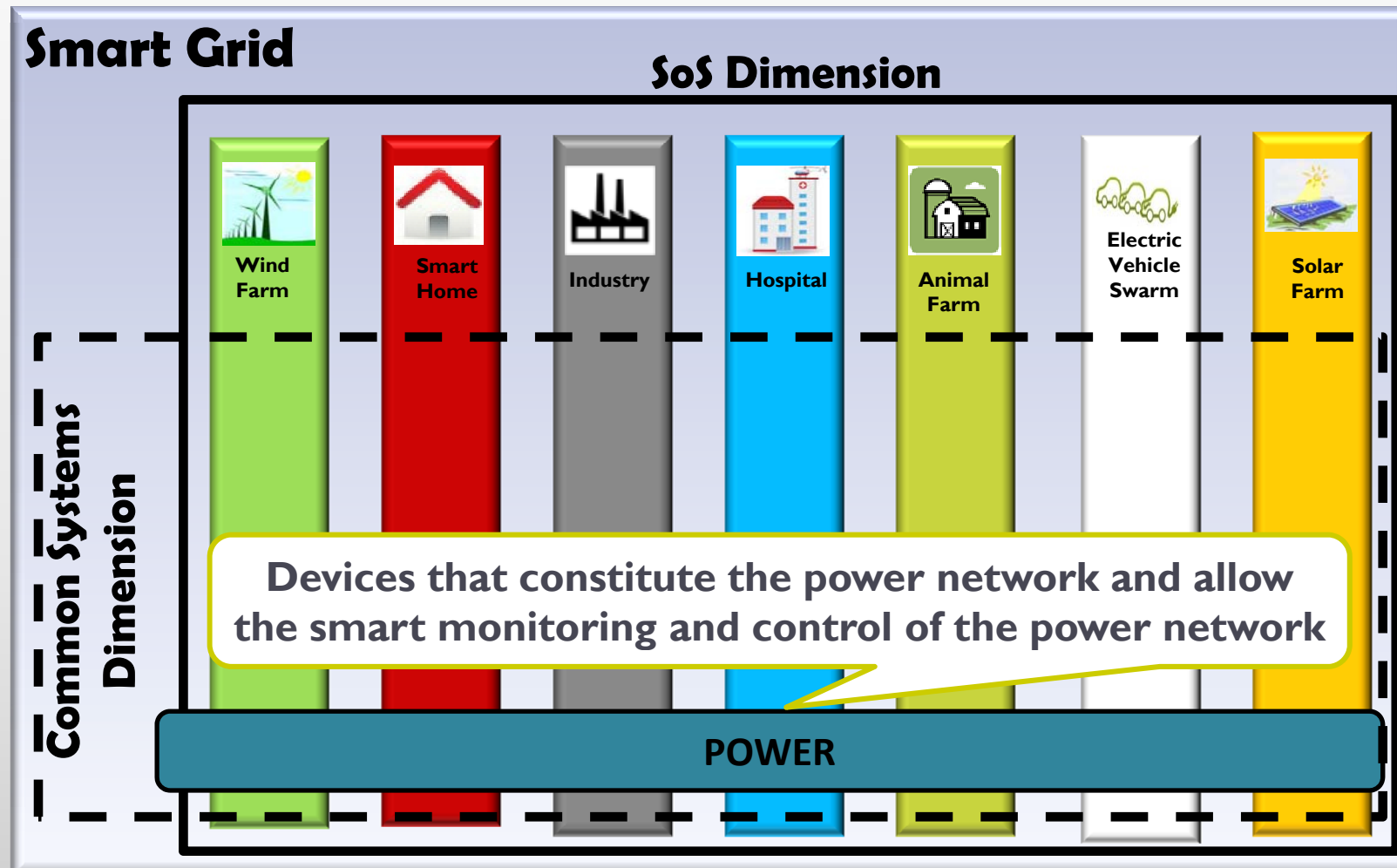
- ☐ Management of real time problems in a more efficient and safer way,
- ☐ Facilitating the integration of real time information: to acquire, store, distribute, process, and analyze data in real-time
- ☐ Connecting and disconnecting resources
- ☐ Smart metering and real time monitoring

► Data interoperability

- ☐ Huge amount of data from the volume of the data collected from smart meters and other devices connected to the power network, which are from different vendors

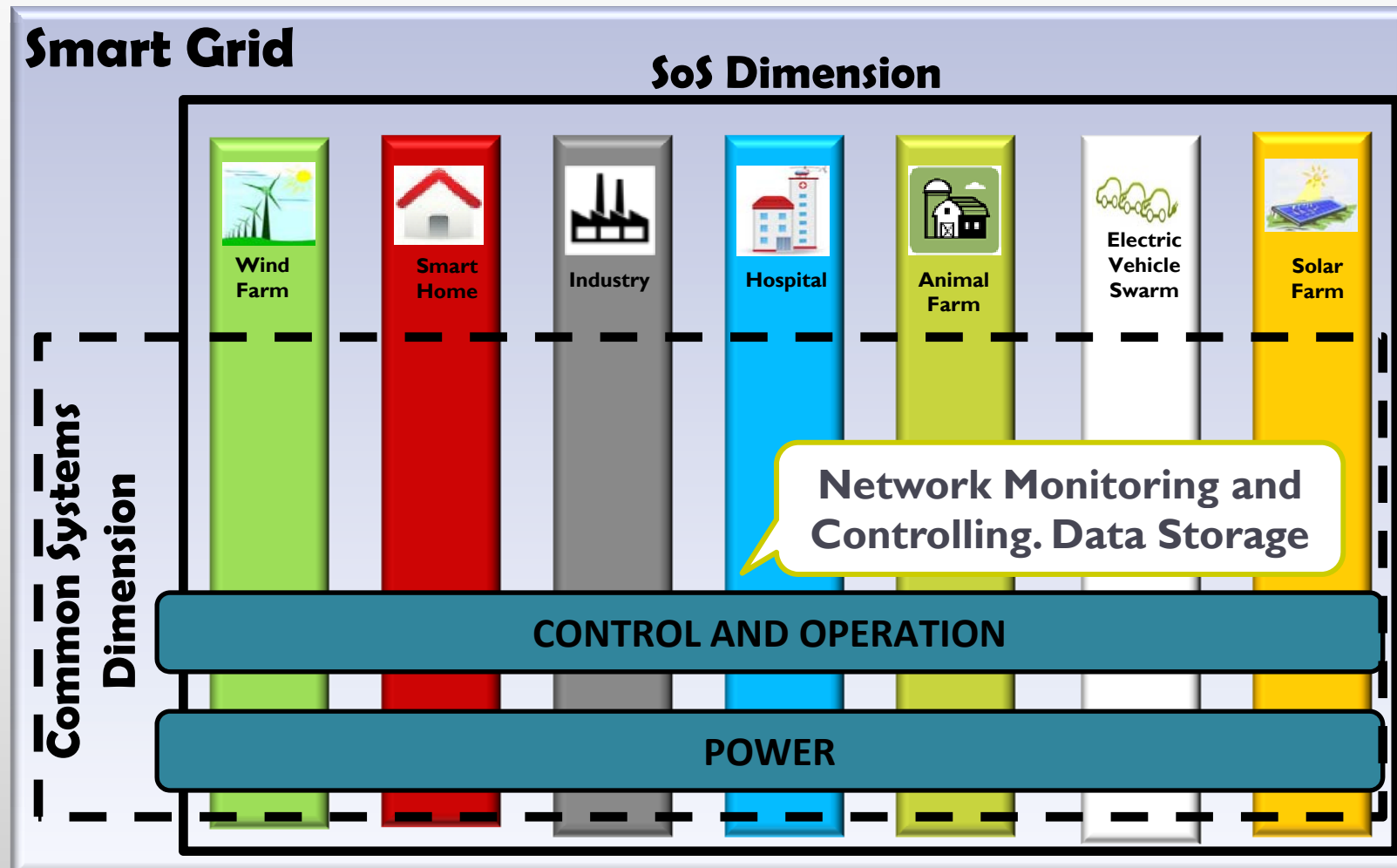
Large-Scale Smart Grids as SoS

► Architectural Framework: **Dimensions**



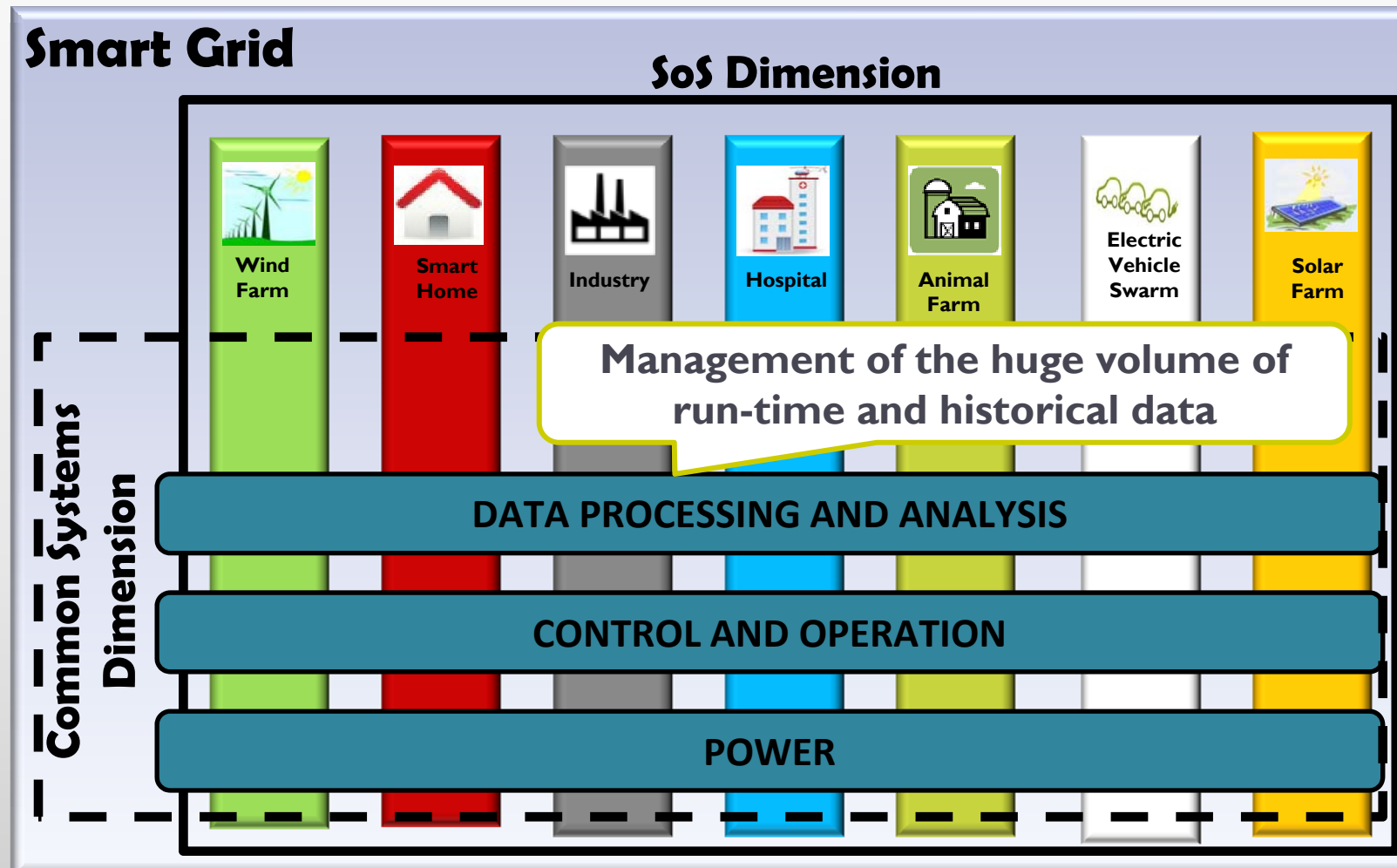
Large-Scale Smart Grids as SoS

► Architectural Framework: **Dimensions**



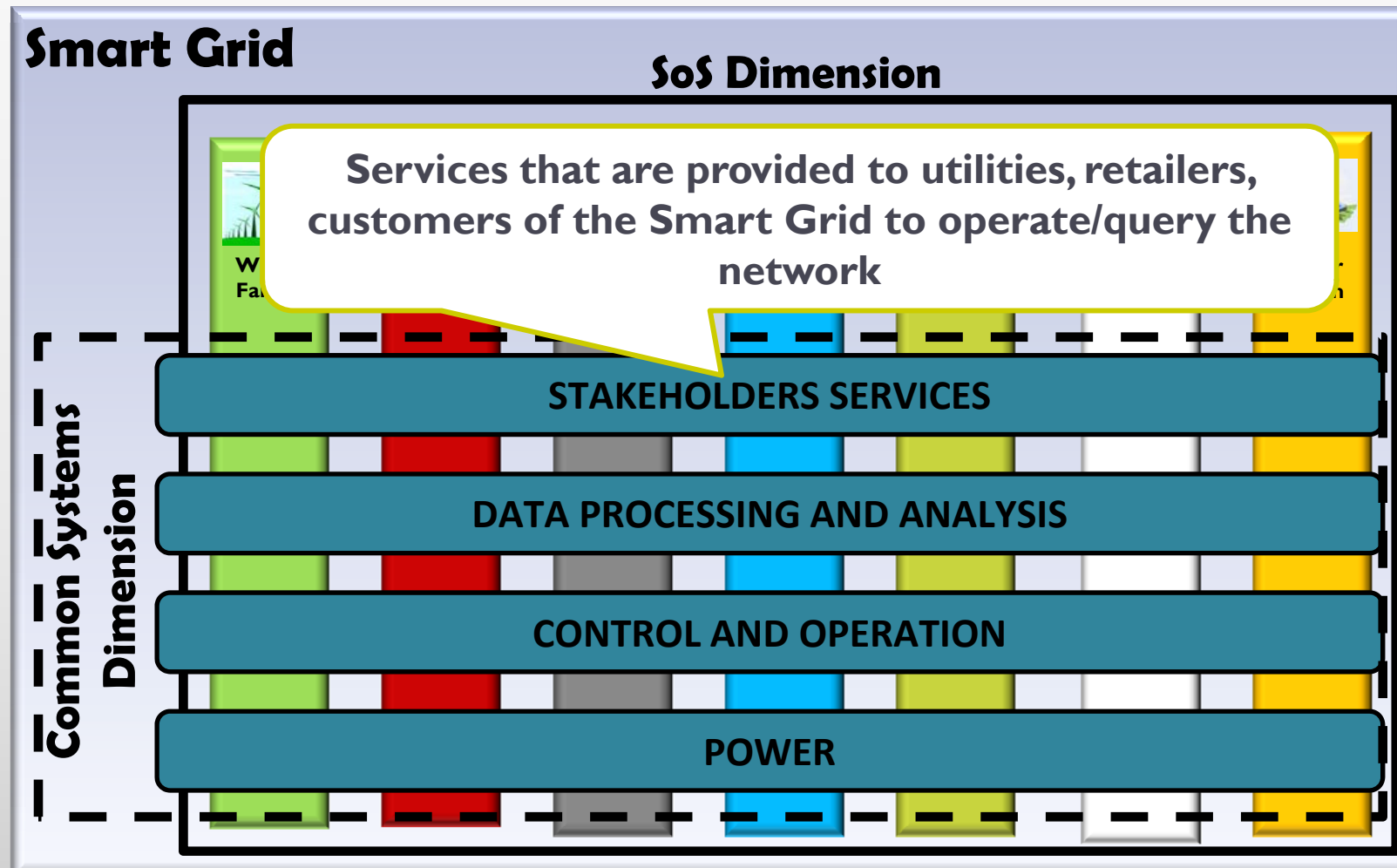
Large-Scale Smart Grids as SoS

► Architectural Framework: **Dimensions**



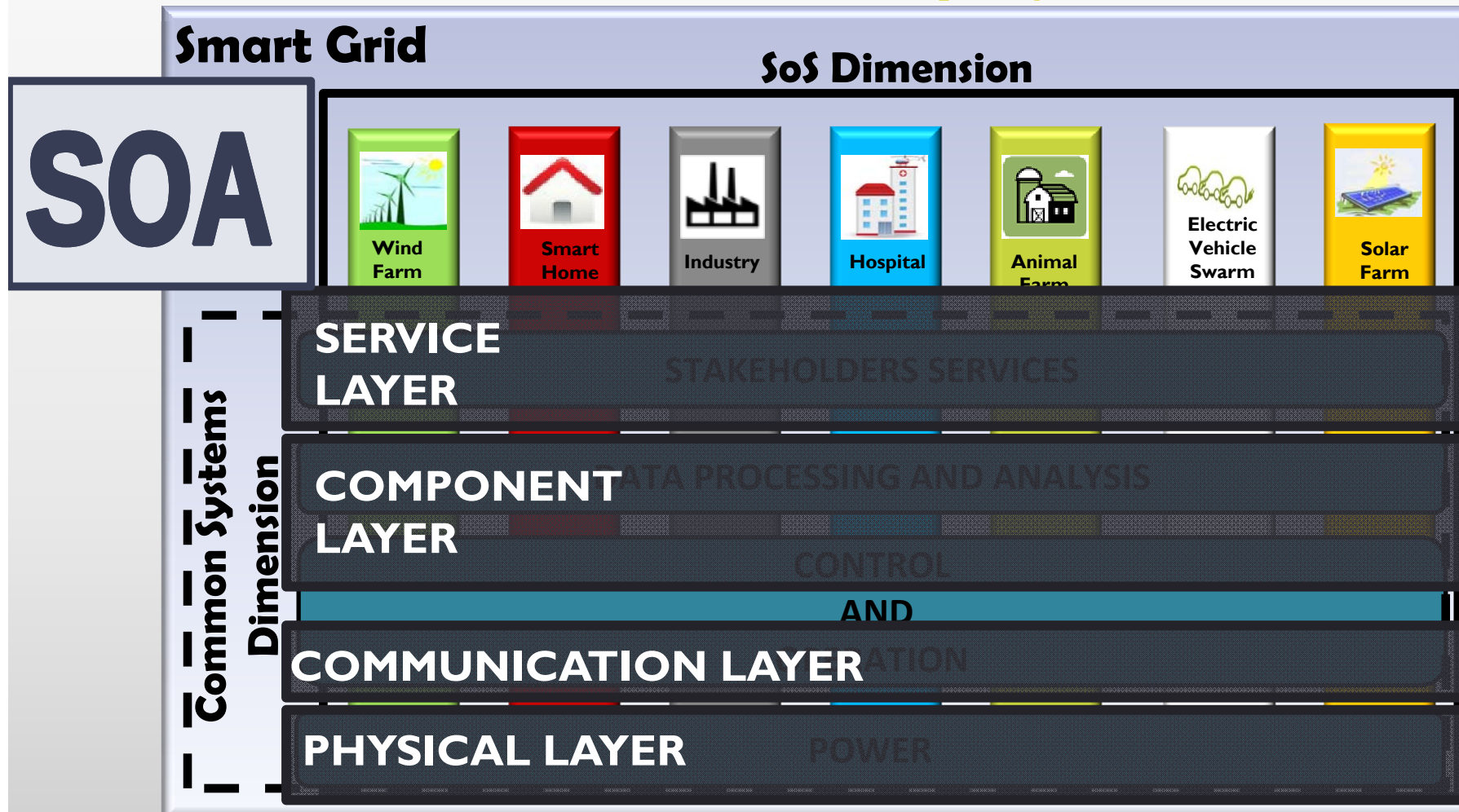
Large-Scale Smart Grids as SoS

► Architectural Framework: **Dimensions**



Large-Scale Smart Grids as SoS

► Architectural Framework: **Deployment**





Large-Scale Smart Grids as SoS

► Architectural Framework: **Deployment**

- Management, access and analysis of a *huge variety of information*
 - System: Data Processing and Analysis
 - Layer: Component
 - Technologies: Massive storage solutions → Big Data
 - Clustering technologies for batch data processing : Hadoop
 - Integration with the SQL and NonSQL paradigms
- *Interoperability of data*
 - System: Operation
 - Layer: Communications
 - Standards: Common Information Model (CIM), IEC 61850
- *Real-Time*
 - System: Operation
 - Layer: Communications
 - Technologies: Extreme processing XTPP which is based on the Publish/Subscribe paradigm with DDS (Data Distribution Service) and CEP (Complex Event Processing)



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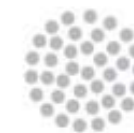
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Conclusions

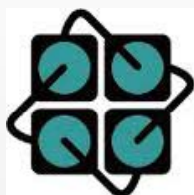
- ▶ From the experience in the projects ENERGOS, NEMO & CODED and IMPONET:
 - ▶ Large-Scale Smart Grids have been defined as a *SoS of SoS*
 - ▶ It has been defined based *Architectural Framework* to deploy the software architectures of Large-Scale Smart Grids.
 - ▶ **Two Dimensions:**
 - Organizational perspective: SoS that compose the Smart Grid
 - Functional perspective: System that provide the common functionality of the Smart Grids
 - ▶ This architectural framework is based on *SOA + run-time capabilities* and a set of technologies and standards to guarantee the open, decoupling and interoperability requirements of SoS architectures.
 - ▶ This is just a first and small step in a long way



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indra



energos



NEMO&CODED

NEtworked Monitoring & Control, Diagnostic for Electrical Distribution



IMPONET

Intelligent Monitoring of
POwer NETworks

**Thank you very much for your
attention
Questions?**

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